

Lab 4 – Vibe Coding Chat History

Python/Pygame Flappy Bird project — Tasks 1 to 4

User

I am working on a Python/Pygame Flappy Bird project for a Vibe Coding lab.

Please inspect the existing project files, especially `game/game_engine.py`, `game/bird.py`, and `game/pipe.py`.

Task 1 from the README is:

"Refine Collision Detection: The bird sometimes clips the edge of a pipe without dying, especially as pipe speed increases. Investigate and enhance collision accuracy so hits are detected consistently."

Analyze the current collision implementation and identify why it can miss collisions.

Then modify the code to use reliable full-bird collision detection against the pipe rectangles instead of checking only the bird's center point.

Requirements:

- Preserve the existing game behavior.
- Do not implement the other README tasks yet.
- Do not add a Game Over screen, replay system, difficulty selection, or sounds yet.
- Make the minimum necessary code changes for Task 1.
- Keep the code clean and readable.
- Explain which files you changed and why.
- Do not change unrelated functionality.

GitHub Copilot

You need to set up GitHub Copilot and be signed in to use Chat.

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GitHub Copilot

I'll trace the current update/collision path in the three game modules, then make the smallest local change and run a focused check of the edited code.

Read []()

Read memory

Read []()

Read []()

Read []()

The local cause is clear: the engine tests whether each pipe contains only the bird's center, so contact at the bird's visible edge is ignored. I'll replace that point test with rectangle overlap using the existing `Bird.rect()` and pipe rectangles; the current 4-pixel pipe movement and 30-pixel bird width also make a focused overlap check a useful way to confirm the edge case.

Ran terminal command: Get-Content

'C:\Users\Del\vscode\extensions\ms-python.vscode-pylance-2026.4.1\skills\python-fact-grounded-coding\SKILL.md'

Ran `pylanceRunCodeSnippet`

Completed with input: {

"workspaceRoot": "c:\\Users\\Del\\40_flappy-bird",

"workingDirectory": "c:\\Users\\Del\\40_flappy-bird",

"codeSnippet": "import pygame\n\npipe = pygame.Rect(100, 0, 70, 120)\nbird_center = (95, 115)\nbird_rect = pygame.Rect(80, 100, 30, 30)\nprint({'center_point_hits': pipe.collidepoint(bird_center), 'full_bird_hits': pipe.colliderect(bird_rect)})"

}

The runtime check confirms the miss: with the bird rectangle overlapping a pipe, `collidepoint(center)` is false while `colliderect(bird\_rect)` is true. I'm changing only the engine's collision predicate, reusing the bird rectangle for both pipe sections; movement, scoring, and game-over handling stay as they are.

Generating patch (20 lines) in []()

Ran `pylanceRunCodeSnippet`

Completed with input: {

"workspaceRoot": "c:\\Users\\Del\\40_flappy-bird",

"workingDirectory": "c:\\Users\\Del\\40_flappy-bird",

"codeSnippet": "import pygame\n\nfrom game.game_engine import GameEngine\n\nfrom game.pipe import Pipe\n\npygame.font.init()\nengine = GameEngine(400, 300)\nengine.bird.x = 95\nengine.bird.y = 114\npipe = Pipe(100, 300, speed=0)\npipe.gap_y = 120\nengine.pipes = [pipe]\nengine.update()\nprint({'bird_center': engine.bird.center(), 'bird_rect': engine.bird.rect(), 'game_over': engine.game_over})\nassert engine.game_over"

}

The first engine-level check hit an unrelated setup constraint: pipe gap randomization needs a screen taller than the default gap plus margins. I'm rerunning the same focused collision scenario at a valid screen height; no code change is needed for that setup failure.

Ran `pylanceRunCodeSnippet`

Completed with input: {

"workspaceRoot": "c:\\Users\\Del\\40_flappy-bird",

"workingDirectory": "c:\\Users\\Del\\40_flappy-bird",

"codeSnippet": "import pygame\n\nfrom game.game_engine import GameEngine\n\nfrom game.pipe import Pipe\n\npygame.font.init()\nengine = GameEngine(400, 500)\nengine.bird.x = 95\nengine.bird.y = 114\npipe = Pipe(100, 500, speed=0)\npipe.gap_y = 120\nengine.pipes = [pipe]\nengine.update()\nprint({'bird_center': engine.bird.center(), 'bird_rect': engine.bird.rect(), 'game_over': engine.game_over})\nassert engine.game_over"

}

Checked [](), no problems found

Updated `game\_engine.py` to check the bird's full rectangle against both pipe rectangles. The old check tested only the bird's center, so it missed collisions when the bird's edge overlapped a pipe but its center remained outside.

The focused runtime check confirmed that edge overlap now sets `game\_over`; the edited file also has no reported errors. No other files or gameplay behavior were changed.

Made changes.

User

Now implement Task 2 from the README: Add a proper Game Over screen.

Requirements:

- When the bird collides with a pipe, the ground, or the ceiling, the game must enter a game-over state.
- Display a clear graphical "GAME OVER" message inside the Pygame window.
- Display the final score on the Game Over screen.
- Stop normal gameplay updates after game over.
- Keep the existing score display during gameplay.
- Do not add the replay button or difficulty selection yet; those belong to Task 3.
- Do not add sound effects yet; those belong to Task 4.
- Keep the collision detection changes from Task 1.
- Preserve the existing game controls and gameplay.
- Make the minimum necessary changes.
- Keep the code clean and readable.
- Explain which files you changed and why.
- Test the game after making the changes and make sure there are no Python errors.

GitHub Copilot

I'll follow the existing render/update flow so the game-over screen fits the current loop and freezes gameplay without changing controls. I'll inspect the engine and main loop, then make the smallest change and run a focused game-state check plus a Python launch/syntax check.

Read []()

Read []()

Read memory

Read []()

The engine already marks pipe, ceiling, and ground hits as game over, and `update()` freezes later frames. The missing piece is purely presentation: `render()` only prints a console message. I'll add a centered, readable overlay with the final score, remove that print-only behavior, and return immediately on a pipe hit so no remaining gameplay work runs that frame.

Generating patch (42 lines) in []()

Ran `pylanceRunCodeSnippet`

Completed with input: {

"workspaceRoot": "c:\\Users\\Dell\\40_flappy-bird",

"workingDirectory": "c:\\Users\\Dell\\40_flappy-bird",

```
"codeSnippet": "import pygame\nfrom game.game_engine import GameEngine\nfrom game.pipe import Pipe\n\npygame.font.init()\nengine = GameEngine(500, 700)\nengine.bird.x = 130\nengine.bird.y = 350\nengine.score = 3\nncolliding_pipe = Pipe(140, 700, speed=0)\nncolliding_pipe.gap_y = 360\nuntouched_pipe = Pipe(300, 700, speed=2)\nengine.pipes = [colliding_pipe, untouched_pipe]\nuntouched_x = untouched_pipe.x\nengine.update()\nassert engine.game_over\nassert untouched_pipe.x == untouched_x\nbird_state = (engine.bird.y, engine.bird.velocity, engine.score)\nengine.handle_event(pygame.event.Event(pygame.KEYDOWN, key=pygame.K_SPACE))\nengine.update()\nassert (engine.bird.y, engine.bird.velocity, engine.score) == bird_state\nscreen = pygame.Surface((500, 700))\nscreen.fill((135, 206, 235))\nengine.render(screen)\nassert screen.get_at((0, 0)) != (135, 206, 235, 255)\nfor y in (14, 686):\n    boundary_engine = GameEngine(500, 700)\n    boundary_engine.bird.y = y\n    boundary_engine.update()\n    assert boundary_engine.game_over\n    print(f'pipe, ceiling, ground, frozen-state, event-ignore, and render checks passed')"
```

}

Checked [](), no problems found

The focused checks now pass for all three death causes, rendering the overlay, and keeping the bird, pipes, score, and flap state frozen after game over. I'm doing one final headless pass through the actual `main()` loop to catch integration errors.

Ran `pylanceRunCodeSnippet`

Completed with input: {

```
"workspaceRoot": "c:\\Users\\Dell\\40_flappy-bird",
```

```
"workingDirectory": "c:\\Users\\Dell\\40_flappy-bird",
```

```
"codeSnippet": "import os\nos.environ[\"SDL_VIDEODRIVER\"] = \"dummy\"\nimport pygame\nimport\nmain\npygame.event.post(pygame.event.Event(pygame.QUIT))\nmain.main()\nprint(\"main loop smoke test passed\")"\n}
```

Implemented Task 2 in `game\_engine.py`. Game over now displays a centered "GAME OVER" message and final score over a dimmed game view; the in-game score remains visible during play. Pipe collisions stop the current update immediately, and further gameplay updates and flap inputs are ignored after game over. Existing ceiling and ground detection remain intact, along with Task 1's full-bird collision checks.

Verified pipe, ceiling, and ground collisions, the frozen game-over state, overlay rendering, and a headless run through `main()`. No Python errors were reported.

Made changes.

User

Now implement Task 3 from the README: Add replay functionality and difficulty selection.

Requirements:

- After Game Over, display a clear menu with options to play again or exit.
- Allow the player to select Easy, Medium, or Hard difficulty.
- The difficulty must have a meaningful gameplay effect, such as changing pipe movement speed and/or the pipe gap.
- The selected difficulty should be used when starting or restarting the game.
- Allow the player to restart without closing and rerunning the Python program.
- Preserve Task 1 full-bird collision detection.
- Preserve Task 2's Game Over screen and final score.
- Do not add sound effects yet; sounds are Task 4.
- Keep the existing gameplay controls.
- Make the minimum necessary changes and do not modify unrelated functionality.
- Keep the code clean and readable.
- Explain which files were changed and why.
- Test all three difficulty modes and the replay flow to make sure there are no Python errors.

GitHub Copilot

The replay/menu behavior belongs in `GameEngine`, while exiting from that menu needs one small hook into `main()`'s loop. I'll start new games at Medium, provide Easy/Medium/Hard selection after game over, apply the chosen speed and gap on Enter, and keep the existing in-game flap controls unchanged.

Apply Patch

I'll keep replay inside the existing engine instance: add three speed/gap presets, a keyboard-selectable game-over menu, and an engine reset for "Play Again." I'm checking the pipe constructor and main loop boundary now, since those determine the smallest clean way to apply the chosen difficulty and exit from the menu.

Read []()

Read

Read []()

The pipe API already accepts both `gap` and `speed`, so no changes to `pipe.py` are needed. I'll add Easy/Medium/Hard presets while retaining Medium's current behavior, draw a post-game menu with keyboard selection, and have `main.py` honor the menu's Exit choice; restarting will rebuild the bird and pipes with the selected preset.

Generating patch (122 lines) in []()

Ran `pylanceRunCodeSnippet`

Completed with input: {

"workspaceRoot": "c:\\Users\\Dell\\40_flappy-bird",

"workingDirectory": "c:\\Users\\Dell\\40_flappy-bird",

```
"codeSnippet": "import pygame\nfrom game.game_engine import GameEngine\n\npygame.font.init()\nengine = GameEngine(500, 700)\nassert (engine.difficulty, engine.pipe_speed, engine.pipes[0].gap) == ("Medium", 4, 150)\nfor key, expected in ((pygame.K_LEFT, ("Easy", 3, 190)), (pygame.K_RIGHT, ("Medium", 4, 150)), (pygame.K_RIGHT, ("Hard", 6, 120))):\n    engine.game_over = True\n    engine.handle_event(pygame.event.Event(pygame.KEYDOWN, key=key))\n    engine.handle_event(pygame.event.Event(pygame.KEYDOWN, key=pygame.K_RETURN))\n    assert (engine.difficulty, engine.pipe_speed, engine.pipes[0].gap) == expected\n    engine._spawn_timer = engine.pipe_interval - 1\n    engine.update()\n    assert engine.pipes[-1].gap == expected[2]\n    assert engine.pipes[-1].speed == expected[1]\n    engine.score = 7\n    engine.game_over = True\n    engine.handle_event(pygame.event.Event(pygame.KEYDOWN, key=pygame.K_r))\n    assert not engine.game_over and engine.score == 0\n    assert (engine.difficulty, engine.pipe_speed, engine.pipes[0].gap) == ("Hard", 6, 120)\n    engine.game_over = True\n    engine.handle_event(pygame.event.Event(pygame.KEYDOWN, key=pygame.K_DOWN))\n    engine.handle_event(pygame.event.Event(pygame.KEYDOWN, key=pygame.K_RETURN))\n    assert engine.exit_requested\n    screen = pygame.Surface((500, 700))\n    engine.game_over = True\n    engine.render(screen)\n    print("all difficulty presets, spawned pipes, replay, exit selection, and menu rendering passed")"
```

Checked [], [], no problems found

The engine checks pass across Easy, Medium, and Hard, including newly spawned pipes, both replay paths, and choosing Exit. I'm wiring that Exit choice into the app loop now and making the selected gap an explicit engine setting so every spawned pipe uses the chosen preset.

Generating patch (33 lines) in [], []

Ran `pylanceRunCodeSnippet`

Completed with input: {

"workspaceRoot": "c:\\Users\\Dell\\40_flappy-bird",

"workingDirectory": "c:\\Users\\Dell\\40_flappy-bird",

```
"codeSnippet": "import pygame\nfrom game.game_engine import GameEngine\nfrom game.pipe import Pipe\n\npygame.font.init()\nengine = GameEngine(500, 700)\nassert (engine.difficulty, engine.pipe_speed, engine.pipe_gap) == ("Medium", 4, 150)\nfor key, expected in ((pygame.K_LEFT, ("Easy", 3, 190)), (pygame.K_RIGHT, ("Medium", 4, 150)), (pygame.K_RIGHT, ("Hard", 6, 120))):\n    engine.game_over = True\n    engine.handle_event(pygame.event.Event(pygame.KEYDOWN, key=key))\n    engine.handle_event(pygame.event.Event(pygame.KEYDOWN, key=pygame.K_RETURN))\n    assert (engine.difficulty, engine.pipe_speed, engine.pipe_gap) == expected\n    engine._spawn_timer = engine.pipe_interval - 1\n    engine.update()\n    assert engine.pipes[-1].gap == expected[2] and engine.pipes[-1].speed == expected[1]\n    engine.score = 7\n    engine.game_over = True\n    engine.handle_event(pygame.event.Event(pygame.KEYDOWN, key=pygame.K_r))\n    assert not engine.game_over and engine.score == 0\n    assert (engine.difficulty, engine.pipe_speed, engine.pipe_gap) == ("Hard", 6, 120)\n    engine.game_over = True\n    engine.handle_event(pygame.event.Event(pygame.KEYDOWN, key=pygame.K_DOWN))\n    engine.handle_event(pygame.event.Event(pygame.KEYDOWN, key=pygame.K_RETURN))\n    assert engine.exit_requested\n    collision_engine = GameEngine(500, 700)\n    collision_engine.bird.x = 130\n    collision_engine.bird.y = 350\n    pipe = Pipe(140, 700, speed=0)\n    pipe.gap_y = 360\n    collision_engine.pipes = [pipe]\n    collision_engine.update()\n    assert collision_engine.game_over\n    screen = pygame.Surface((500, 700))\n    collision_engine.render(screen)\n    print("difficulty, replay, exit, Task 1 collision, and game-over rendering checks passed")"
```

Ran `pylanceRunCodeSnippet`

Completed with input: {

"workspaceRoot": "c:\\Users\\Dell\\40_flappy-bird",

"workingDirectory": "c:\\Users\\Dell\\40_flappy-bird",

```
"codeSnippet": "import os\nos.environ["SDL_VIDEODRIVER"] = "dummy"\n\nimport pygame\nimport main\n\nmain.engine.game_over = True\nmain.engine.menu_selection = 1\npygame.event.post(pygame.event.Event(pygame.KEYDOWN, key=pygame.K_RETURN))\nmain.main()\nassert main.engine.exit_requested\nprint("main loop menu-exit smoke test passed")"
```

Checked [], [], no problems found

Implemented replay and difficulty selection in `game\_engine.py` and `main.py`. The game-over menu lets players choose Easy, Medium, or Hard, then select Play Again or Exit. Left/Right (or A/D) changes difficulty; Up/Down (or W/S) changes the menu selection; Enter confirms. `R` replays immediately with the selected difficulty, and Escape exits.

Easy uses speed 3 and gap 190; Medium retains the existing speed 4 and gap 150; Hard uses speed 6 and gap 120. Restart resets the score and game state, and all new pipes use the selected settings. Task 1 collision detection and Task 2's final score screen remain intact.

Tested all difficulty settings, spawned-pipe settings, replay, exit, collision detection, and the main-loop exit flow. No Python errors were reported.

Made changes.

User

Now implement Task 4 from the README: Add sound feedback to the Flappy Bird game.

Requirements:

- Add a flap sound whenever the bird flaps.
- Add a score/pass sound whenever the player successfully passes a pipe and the score increases.
- Add a death sound whenever the bird dies from a pipe, ground, or ceiling collision.
- Use appropriate local sound files in the project.
- If sound assets need to be added, organize them cleanly in an assets or sounds folder.
- Load sounds once rather than recreating them every frame.
- Keep the existing gameplay controls unchanged.
- Preserve Task 1 full-bird collision detection.
- Preserve Task 2 Game Over screen and final score.
- Preserve Task 3 replay functionality and Easy/Medium/Hard difficulty selection.
- Do not change unrelated gameplay behavior.
- Make the minimum necessary changes.
- Keep the code clean and readable.
- Explain which files and assets were added or changed and why.
- Test the game and verify that flap, score, and death sounds trigger at the correct events.
- Make sure there are no Python errors.

GitHub Copilot

I'll check how flap, scoring, and each death condition are currently handled, plus whether the project already has suitable audio assets or mixer setup. Then I'll add small local WAV effects and hook playback into those existing events without changing controls or replay behavior.

Searched for files matching `\*\*/\*.{wav,ogg,mp3,flac}`, no matches

Searched for files matching `\*\*/\*`, 9 matches

Read []()

Read []()

Read []()

There are no existing sound assets, and the event hooks are all in `game\_engine.py`. I'll add three short local WAV effects (flap chirp, score chime, descending death tone), then load them once when the engine is created and play them only at the corresponding events.

Ran `pylanceRunCodeSnippet`

Completed with input: {

"workspaceRoot": "c:\\Users\\Dell\\40_flappy-bird",

"workingDirectory": "c:\\Users\\Dell\\40_flappy-bird",

```
"codeSnippet": "from pathlib import Path\nimport math\nimport struct\nimport wave\n\nsample_rate = 22050\nsound_dir = Path(\"assets/sounds\")\nsound_dir.mkdir(parents=True, exist_ok=True)\n\ndef write_sound(filename, duration, kind):\n    frame_count = int(sample_rate * duration)\n    frames = bytearray()\n    for frame in range(frame_count):\n        time = frame / sample_rate\n        if kind == \"flap\":\n            frequency = 1050 - 420 * (time / duration)\n            envelope = min(1.0, time / 0.008) * math.exp(-time * 12)\n            value = math.sin(2 * math.pi * frequency * time)\n        elif kind == \"score\":\n            note_time = time if time < 0.12 else time - 0.12\n            frequency = 880 if time < 0.12 else 1320\n            envelope = min(1.0, note_time / 0.006) * math.exp(-note_time * 10)\n            value = math.sin(2 * math.pi * frequency * time)\n        else:\n            frequency = 360 - 250 * (time / duration)\n            envelope = min(1.0, time / 0.01) * math.exp(-time * 3.5)\n            value = 0.8 * math.sin(2 * math.pi * frequency * time) + 0.2 * math.sin(4 * math.pi * frequency * time)\n        sample = int(0.45 * 32767 * envelope * value)\n        frames.extend(struct.pack(\"<h\",
```

```
sample))\n path = sound_dir / filename\n with wave.open(str(path), 'wb') as wav_file:\n wav_file.setnchannels(1)\n wav_file.setsampwidth(2)\n wav_file.setframerate(sample_rate)\n wav_file.writeframes(frames)\n return path, len(frames)\n\nfor name, duration, kind in (('flap.wav', 0.12, 'flap'), ('score.wav', 0.25, 'score'), ('death.wav', 0.38, 'death')):\n path, size = write_sound(name, duration, kind)\n with wave.open(str(path), 'rb') as wav_file:\n assert wav_file.getnframes() > 0 and wav_file.getframerate() == sample_rate\n print(f'{path}: {size} bytes')"
```

Generating patch (63 lines) in []()

The three WAV assets are in place, and sound loading is isolated to engine initialization with a no-audio fallback. I've wired the three event sites; I'm now checking playback calls at flap, pass, and all death boundaries, as well as confirming replay doesn't recreate sound objects.

Ran `pylanceRunCodeSnippet`

Completed with input: {

"workspaceRoot": "c:\\Users\\Dell\\40_flappy-bird",

"workingDirectory": "c:\\Users\\Dell\\40_flappy-bird",

```
"codeSnippet": "import os\nos.environ['SDL_AUDIODRIVER'] = 'dummy'\n\nimport pygame\npygame.init()\n\nfrom game.game_engine import GameEngine\n\nfrom game.pipe import Pipe\n\nengine = GameEngine(500, 700)\nassert set(engine.sounds) == {'flap', 'score', 'death'}\n\nplayed = []\n\nclass SoundSpy:\n    def __init__(self, name):\n        self.name = name\n    def play(self):\n        played.append(self.name)\n\nengine.sounds = {name: SoundSpy(name) for name in ('flap', 'score', 'death')}\n\nengine.handle_event(pygame.event.Event(pygame.KEYDOWN, key=pygame.K_SPACE))\nengine.handle_event(pygame.event.Event(pygame.MOUSEBUTTONDOWN, button=1))\n\nassert played == ['flap', 'flap']\n\nengine.bird.y = 350\npipe = Pipe(50, 700, gap=150, speed=0)\npipe.gap_y = 270\nengine.pipes = [pipe]\n\nengine.update()\n\nassert engine.score == 1 and played[-1] == 'score'\n\nfor cause in ('ceiling', 'ground', 'pipe'):\n    death_engine = GameEngine(500, 700)\n    death_engine.sounds = {name: SoundSpy(name) for name in ('flap', 'score', 'death')}\n    if cause == 'ceiling':\n        death_engine.bird.y = 14\n    elif cause == 'ground':\n        death_engine.bird.y = 686\n    else:\n        death_engine.bird.x = 130\n    death_engine.bird.y = 350\n    pipe = Pipe(140, 700, speed=0)\n    pipe.gap_y = 360\n    death_engine.pipes = [pipe]\n    death_engine.update()\n    assert death_engine.game_over\n    assert played[-1] == 'score' or True\n    death_engine._play_sound('death')\n    assert played[-1] == 'death'\n    death_engine.update()\n    assert len(engine.sounds) == 3\n    sound_objects = engine.sounds\n    engine.game_over = True\n    engine.handle_event(pygame.event.Event(pygame.KEYDOWN, key=pygame.K_r))\n    assert engine.sounds is sound_objects\n    print('flap, score, ceiling, ground, pipe death, and replay sound checks passed')
```

Checked , [](), no problems found

The first runtime pass confirmed flap and scoring playback, but I spotted that its death assertion didn't strictly isolate the death callback. I'm tightening that check to require exactly one death playback for each collision cause and none on later frozen frames.

Ran `pylanceRunCodeSnippet`

Completed with input: {

"workspaceRoot": "c:\\Users\\Dell\\40_flappy-bird",

"workingDirectory": "c:\\Users\\Dell\\40_flappy-bird",

```
"codeSnippet": "import os\nos.environ['SDL_AUDIODRIVER'] = 'dummy'\n\nimport pygame\npygame.init()\n\nfrom game.game_engine import GameEngine\n\nfrom game.pipe import Pipe\n\nplayed = []\n\nclass SoundSpy:\n    def __init__(self, name):\n        self.name = name\n    def play(self):\n        played.append(self.name)\n\nengine = GameEngine(500, 700)\nassert set(engine.sounds) == {'flap', 'score', 'death'}\n\nassert all(sound.get_length() > 0 for sound in engine.sounds.values())\n\nengine.sounds = {name: SoundSpy(name) for name in ('flap', 'score', 'death')}\n\nengine.handle_event(pygame.event.Event(pygame.KEYDOWN, key=pygame.K_SPACE))\nengine.handle_event(pygame.event.Event(pygame.MOUSEBUTTONDOWN, button=1))\n\nassert played == ['flap', 'flap']\n\nplayed.clear()\n\nengine.bird.y = 350\npipe = Pipe(50, 700, gap=150, speed=0)\npipe.gap_y = 270\nengine.pipes = [pipe]\n\nengine.update()\n\nassert engine.score == 1 and played == ['score']\n\nfor cause in ('ceiling', 'ground', 'pipe'):\n    played.clear()\n    death_engine = GameEngine(500, 700)\n    death_engine.sounds = {name: SoundSpy(name) for name in ('flap', 'score', 'death')}\n    if cause == 'ceiling':\n        death_engine.bird.y = 14\n    elif cause == 'ground':\n        death_engine.bird.y = 686\n    else:\n        death_engine.bird.x = 130\n    death_engine.bird.y = 350\n    pipe = Pipe(140, 700, speed=0)\n    pipe.gap_y = 360\n    death_engine.pipes = [pipe]\n    death_engine.update()\n    assert death_engine.game_over and played == ['death']\n    (cause, played)\n    death_engine.update()\n    assert played == ['death']\n    (cause, played)\n    sound_objects = engine.sounds\n    engine.game_over = True\n    engine.handle_event(pygame.event.Event(pygame.KEYDOWN, key=pygame.K_r))\n    assert engine.sounds is sound_objects\n    print('flap, score, each death cause, no repeated death sound, and replay sound reuse passed')
```

Checked [](), [](), no problems found

The stricter event checks now confirm exact playback: flap for keyboard and mouse input, score only on a successful pass, and one death sound per ceiling, ground, or pipe collision. Replay keeps the loaded sound objects; I'm finishing with an application-loop smoke test in dummy audio/video mode.

Ran `pylanceRunCodeSnippet`

Completed with input: {

"workspaceRoot": "c:\\Users\\Dell\\40_flappy-bird",

"workingDirectory": "c:\\Users\\Dell\\40_flappy-bird",

"codeSnippet": "import os\nos.environ[\"SDL_VIDEODRIVER\"] = \"dummy\"\nos.environ[\"SDL_AUDIODRIVER\"] = \"dummy\"\nimport\npygame\nimport main\nassert set(main.engine.sounds) == {\"flap\", \"score\",\n\"death\"}\npygame.event.post(pygame.event.Event(pygame.QUIT))\nmain.main()\nprint(\"Pygame app loop and sound loading smoke test\npassed\")"

}

Added event-driven audio in `game\_engine.py`. It loads the sounds once when the engine is created and reuses them during replay:

- `flap.wav` plays on Space or mouse flap.
- `score.wav` plays when passing a pipe increases the score.
- `death.wav` plays on pipe, ceiling, or ground collision.

Task 1 collision detection, the Task 2 game-over screen, Task 3 replay and difficulty selection, and existing controls are preserved. No other gameplay behavior was changed.

Verified each sound trigger, confirmed death audio plays only once per collision, and smoke-tested the app loop with dummy audio/video. No Python errors were reported.

Made changes.